

Milind Bansal

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PROFESSIONAL SUMMARY

Data-driven developer with hands-on experience designing systems that capture, structure, and surface operational insights. Strong foundation in SQL, analytics pipelines, and data modelling, complemented by coursework in DVA, Tableau, AI/ML, DBMS, and Finance.

EDUCATION

Newton School of Technology, Rishihood University <i>B.Tech, Artificial Intelligence (Minor: Finance)</i>	2024 – 2028 <i>GPA: 9.2 / 10.0</i>
• Relevant Coursework: Data & Visual Analytics (DVA), Tableau, AI & Machine Learning, Database Management Systems (DBMS), Finance (Minor)	
Shanti Vidya Mandir <i>Intermediate, Class XII</i>	2023 – 2024 <i>80.0%</i>
Delhi Public School <i>Matriculation, Class X</i>	2022 – 2023 <i>93.0%</i>

TECHNICAL SKILLS

Analytics & BI: Tableau, SQL (PostgreSQL, MySQL), NoSQL (MongoDB), NumPy, Pandas, Matplotlib
Data Engineering: Prisma ORM, PostgreSQL schema design, Event-Driven Architecture, RESTful APIs
AI / ML: Python, LangChain, LangGraph, OpenAI API, GenAI, LLMs, Machine Learning
Languages: Python, SQL, TypeScript, JavaScript, HTML, CSS
Tools & Platforms: Git & GitHub, Figma, Excel, Next.js, React, Node.js, Express.js, OAuth 2.0, Unit Testing

PROJECTS

- Beijing PM2.5 Air Quality Analysis** | *Python, Pandas, Tableau* | [GitHub](#) [Dashboard](#) April 2026
- **Project Lead & ETL Lead** on a 6-member capstone: engineered an end-to-end analytics pipeline on **43,824 hourly observations** (UCI Beijing PM2.5 dataset, 2010–2014) — handling datetime synthesis, forward-fill imputation for 2,067 nulls, and feature engineering of AQI category and season columns.
 - Designed and computed a **6-KPI framework** (Annual Mean, WHO Ratio, Hazardous Rate, AQI Distribution) revealing that Beijing’s PM2.5 exceeded WHO limits by **6–7x every year**, with air classified unsafe for over 50% of all recorded hours.
 - Conducted **7 non-parametric statistical tests** (Spearman, Kruskal-Wallis, Mann-Whitney) establishing wind speed as the strongest pollution reducer ($r = -0.35$), seasonal alerts as statistically justified ($H = 105.37$), and rainfall as a non-significant cleaning agent ($p = 0.63$).
 - Produced **11 EDA visualisations** (monthly time series, hourly heatmaps, AQI donut charts) and a dual-view **Tableau dashboard** with executive KPI summaries and operational drill-downs filterable by Year, Season, and AQI Category.
 - Translated analysis into **3 operational recommendations**: winter-priority coal emission curtailment, NW-wind “Clean Window” alerts, and calm-wind emergency industrial caps — projecting a 10–15% reduction in peak PM2.5 exposure.
- Adani Total Gas Ltd. — Financial Statement Analysis & DCF Valuation** | *Excel, Financial Modelling, DCF2025*
- Constructed a **4-year consolidated financial model** (FY2022–FY2025) for Adani Total Gas Ltd., structuring Income Statement, Balance Sheet, and Common Size statements from raw annual report data — enabling trend and cross-sectional analysis across Rs.3,252Cr to Rs.5,457Cr revenue growth.
 - Computed and interpreted **12+ financial ratios** across four dimensions — Liquidity (Current, Quick, Cash), Efficiency (Asset Turnover, DSO, Fixed Asset Turnover), Profitability (Gross/EBIT/Net Margins, ROE, ROA, ROCE) — surfacing declining margin trends and improving asset utilisation over the period.
 - Built a **DCF valuation model** using both FCFF and FCFE frameworks: estimated Cost of Equity via CAPM (13.8%), projected 3-year free cash flows, and calculated Terminal Value using India’s nominal GDP growth rate (8%) — concluding the stock was **OVERVALUED** at Rs.602.6 vs. intrinsic value of Rs.154.7.

- Performed **sensitivity analysis** on a 5x5 grid varying Cost of Equity (10%–17%) and Terminal Growth Rate (7%–9%), quantifying valuation ranges and communicating model uncertainty to non-technical stakeholders.
- Modelled **Expected Growth Rate (g)** using ROIC x Reinvestment Rate, and documented sector-wise applicability of the formula — distinguishing capital-intensive (Gas, Utilities) from knowledge-intensive sectors (IT, FMCG) where the metric systematically underestimates growth.